Qualitative Content Analysis (Mayring)

Advanced Topics: Methodology, Philosophy, and Research Design

1. Foundations

You've conducted 20 interviews and now have 200 pages of transcripts. How do you keep track of it all?

Qualitative content analysis according to Mayring is the Swiss army knife of text analysis: systematic, rule-based, yet still open to surprises.

Developed by Philipp Mayring (1983, continuously updated).

Goal: Systematic, rule-guided analysis of text material while maintaining openness and traceability.

Object: Any form of fixed communication (interview transcripts, newspaper articles, laws, diaries, images).

2. Three Basic Forms

Summarization: Material paraphrased, generalized, reduced. Goal: essence in brief.

Result: Categorisation system + abstract

Explication: Unclear text passages clarified with additional material.

Two main sub forms: narrow contextual analysis (immediate textual context) vs. broad contextual analysis (additional sources).

Structuring: Specific aspects extracted from the material. Several sub-types:

- Content-based structuring: extracting themes and content.
- Formal structuring: organising by text components (e.g. patterns of argumentation).
- Typological structuring: identifying extremes and special cases.
- Scaling structuring: classifying material on a scale (quasi-quantitative).

3. Procedure Model (E.g. Summarization)

Step 1: Define material (which texts, which parts?).

Step 2: Determine analysis direction.

Step 3: Paraphrasing: text into own concise words.

Step 4: Generalization: paraphrases at uniform abstraction level.

Step 5: Reduction I: delete/merge identical paraphrases.

Step 6: Reduction II: bundle central statements, form new categories.

Step 7: Back-check: verify against original material.

The process model is defined and documented in advance. It is a cyclical process (pilot phase → review).

4. Category System

Heart of content analysis is the category system. It is separated into *main categories, subcategories, anchor examples*.

Inductive category formation: Form categories inductively from the material.
Separate the material into categories or literally cut each category out of the material.
This is a step-by-step generalization that is an iterative approach.
The problem is mostly personal experience and expertise.
You need to be able to answer the question: When is each category 'saturated' or fulfilled?

Deductive category formation: Categories derived from theory and applied to the material.
You start with a theory and some given hypotheses.
When you perform such an analysis you normally already have a theory and given hypotheses as you would not just random sample interview without any intend and start a study.
Consequently, structure your content based on those assumptions and derive categories for coding based on that.

Combined approach: Start with a deductive approach (based on guidelines/theory), then expand upon it inductively.

Coding guide per category: Per category a definition, anchor example, and a coding rule is deducted and applied to the sample.

5. Reliability and Validity

Reliability: Assesses the consistency and stability of results when the test is administered repeatedly.

Inter-coder reliability: Two independent coders. Agreement as Cohen's kappa (kappa > 0.7 acceptable). If the coding system is well-designed, it can achieve high scores as a quality criterion for a meaningful study.

Intra-coder reliability: The same encoder again after a certain amount of time.

Validity: Does the method measure whether it captures what it is supposed to capture? Validation through communication with respondents, triangulation (multiple methods/data types), and a rule-based approach.

Criterion validity: Measures how well the results of an analysis correspond with an external criterion.

When the results of a content analysis are compared with the results of another, already validated method.

Construct validity: Refers to whether the categories and concepts used measure what they are intended to measure. That means derived categories are theoretically sound and the underlying concepts are comprehensible.

Content validity: Assesses whether the categories and items used cover all relevant aspects of the topic under investigation. In case all key dimensions of a topic in order to obtain a comprehensive picture are stated clearly one can assume for validity of the content.

Documenting the process/ Transparency: Every step is documented (checklists, reports). Disclosure of all aspects of the research process to ensure transparency.

6. Software

MAXQDA: Commercial, widely used. Coding, memos, visualisations, category-based analysis.

QCAmap: Open-source web application designed specifically for Mayring content analysis. Automatic checklists.

ATLAS.ti: Similar to MAXQDA. Network analyses.

NVivo: Widely used in English-based research.

f4analyse: Free and easy. Combined with f4transkript.

7. Analyse-Workflow: Inhaltsanalyse nach Mayring

Step 1 – **Selection of material**: Determining which sections of text are to be analysed. Defining the corpus.

Step 2 – **Focus of the analysis**: What do I want to know? What question is guiding the analysis?

Step 3 – **Developing a classification system**: either inductively (based on the material) or deductively (based on theory). Coding guide with definitions, illustrative examples and coding rules.

Step 4 – **Pilot phase**: Test the categorisation system on 10–20% of the material. Revise. Conduct a second pilot round if necessary.

Step 5 – **Main coding**: Code all the material using the category system. If there are multiple coders: Calculate inter-coder reliability (Cohen's Kappa > 0.7).

Step 6 – **Analysis**: Counting frequencies (in quantitative content analysis). Identifying patterns and relationships between categories.

Step 7 – **Interpretation**: Interpret the findings in terms of content. Support categories with quotations. Relate back to the research question and theory.